

Data Collection and Preprocessing Phase

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Maximum Marks	2 Marks

Data Collection Plan and Raw Data Sources: Productivity Prediction Web App

Saurabh Yadav

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Introduction

The Productivity Prediction Web App employs machine learning to forecast garment worker productivity, utilizing the "Productivity Prediction of Garment Employees" dataset. This document outlines the data collection plan and raw data source identification to ensure data integrity for training Linear Regression ($R^2 = 0.197$), Random Forest ($R^2 = 0.547$), and XGBoost ($R^2 = 0.482$) models. By curating high-quality data, the project supports accurate predictive modeling and operational decision-making in the garment industry.

Data Collection Plan

The data collection strategy focuses on acquiring a reliable, publicly available dataset suitable for machine learning analysis. The plan includes:

1. Data Identification:

- Objective: Source a dataset with features relevant to garment worker productivity (e.g., team size, SMV, overtime).
- Criteria: Dataset must include a continuous target variable ('actual_productivity'), diverse input features, and sufficient records (>500).
- Source Type: Public repository (Kaggle) for accessibility and ethical use.

2. Data Acquisition:

- Method: Download the CSV file from Kaggle.

- Timeline: Completed in January 2025 (project initiation).

- Tools: Web browser for download; Pandas for loading and inspection.

3. Data Organization:

- Storage: Save `garments\_worker\_productivity.csv` in the project directory.
- Structure: Maintain raw dataset in CSV format; create processed versions post-preprocessing.
- Documentation: Log dataset metadata (e.g., features, size) in project notes.

4. Data Validation:

- Initial Checks: Verify record count, feature types, and missing values using Pandas.
- Quality Assessment: Identify issues like missing data or inconsistencies (e.g., misspellings).
- Validation Tools: Python scripts with Pandas and NumPy for summary statistics and null checks.

5. Ethical Considerations:

- Ensure dataset is anonymized and complies with Kaggle's public use license.
- Avoid collecting sensitive or proprietary data.

Raw Data Sources

- Dataset: "Productivity Prediction of Garment Employees"
- Source: Kaggle (<https://www.kaggle.com/datasets/utkarshs/productivity-prediction-of-garment-employees>)
- Format: CSV file (`garments\_worker\_productivity.csv`)
- Size: 1197 records, 15 features
- Features:
 - quarter: Quarter of the year (categorical, e.g., Quarter1–Quarter5)
 - department: Department type (categorical, e.g., sweing, finishing)
 - day: Day of the week (categorical, e.g., Monday–Sunday)
 - team: Team number (numerical, 1–12)
 - targeted_productivity: Target productivity (decimal, 0.35–0.8)
 - smv: Standard Minute Value (decimal, 2.9–50.48)
 - wip: Work in Progress (integer, 0–1591, nullable)

- over_time: Overtime in minutes (integer, 0–25920)
- incentive: Incentive amount (integer, 0–98)
- idle_time: Idle time in minutes (decimal, 0–6.5)
- idle_men: Number of idle workers (integer, 0–40)
- no_of_style_change: Style changes (integer, 0–2)
- no_of_workers: Number of workers (decimal, 2–59)
- date: Observation date (string, e.g., 1/1/2015)
- actual_productivity: Target variable (decimal, 0.2337–0.7548)
- License: Public domain (Kaggle Community Data License)
- Collection Date: January 2025
- Relevance: Features capture key productivity drivers (e.g., SMV, team size), ideal for ML regression tasks.

Data Integrity Measures

- Initial Quality Checks:
 - Verified 1197 records with 15 features using Pandas ``info()``.
 - Confirmed no duplicate records with ``duplicated()``.
 - Identified 506 missing values in ``wip`` (42.3%), primarily for ``finishing`` department.
 - Noted inconsistent ``department`` spellings (``sweing`` vs. ``sewing``).
- Validation Steps:
 - Generated summary statistics (mean, min, max) for numerical features (e.g., ``smv``, ``over_time``) using Pandas ``describe()``.
 - Checked categorical feature uniqueness (e.g., ``quarter``: 5 unique values) with ``value_counts()``.
 - Visualized missing data with a heatmap to confirm ``wip`` as the only nullable feature.
- Integrity Assurance:
 - Stored raw dataset unchanged to preserve original data.
 - Documented issues and resolutions in preprocessing logs.
 - Ensured feature types (e.g., categorical, numerical) align with ML requirements.

Impact on Analysis

- The curated dataset enabled robust training of ML models, with Random Forest achieving the highest R^2 score (0.547), followed by XGBoost (0.482) and Linear Regression (0.197).
- Identifying `wip` missing values informed mean imputation, preserving data for `sewing` department predictions.
- Noting `department` inconsistencies led to standardization, ensuring model consistency.
- Deriving `month` from `date` captured temporal trends, enhancing model performance.
- The data collection plan, executed with Python and Pandas, laid the foundation for the app's predictive accuracy, supporting garment industry decision-making.

Images

1. Figure 1: Dataset Preview Table

	date	quarter	department	day	team	targeted_productivity	smv	wip	over_time	incentive	idle_time	idle_m
0	1/1/2015	Quarter1	sweing	Thursday	8	0.80	26.16	1108.0	7080	98	0.0	
1	1/1/2015	Quarter1	finishing	Thursday	1	0.75	3.94	NaN	960	0	0.0	
2	1/1/2015	Quarter1	sweing	Thursday	11	0.80	11.41	968.0	3660	50	0.0	
3	1/1/2015	Quarter1	sweing	Thursday	12	0.80	11.41	968.0	3660	50	0.0	
4	1/1/2015	Quarter1	sweing	Thursday	6	0.80	25.90	1170.0	1920	50	0.0	
...	...	...	...	...	...	...	...	...	...	...	...	...
1192	3/11/2015	Quarter2	finishing	Wednesday	10	0.75	2.90	NaN	960	0	0.0	
1193	3/11/2015	Quarter2	finishing	Wednesday	8	0.70	3.90	NaN	960	0	0.0	
1194	3/11/2015	Quarter2	finishing	Wednesday	7	0.65	3.90	NaN	960	0	0.0	
1195	3/11/2015	Quarter2	finishing	Wednesday	9	0.75	2.90	NaN	1800	0	0.0	
1196	3/11/2015	Quarter2	finishing	Wednesday	6	0.70	2.90	NaN	720	0	0.0	

197 rows × 15 columns

	team	targeted_productivity	smv	wip	\
team	1.000000	0.030274	-0.110011	-0.033474	
targeted_productivity	0.030274	1.000000	-0.069489	0.062054	
smv	-0.110011	-0.069489	1.000000	-0.037837	
wip	-0.033474	0.062054	-0.037837	1.000000	
over_time	-0.096737	-0.088557	0.674887	0.022302	
incentive	-0.007674	0.032768	0.032629	0.167210	
idle_time	0.003796	-0.056181	0.056863	-0.026299	
idle_men	0.026974	-0.053818	0.105901	-0.048718	
no_of_style_change	-0.011194	-0.209294	0.315388	-0.072357	
no_of_workers	-0.075113	-0.084288	0.912176	0.030383	
actual_productivity	-0.148753	0.421594	-0.122089	0.131147	

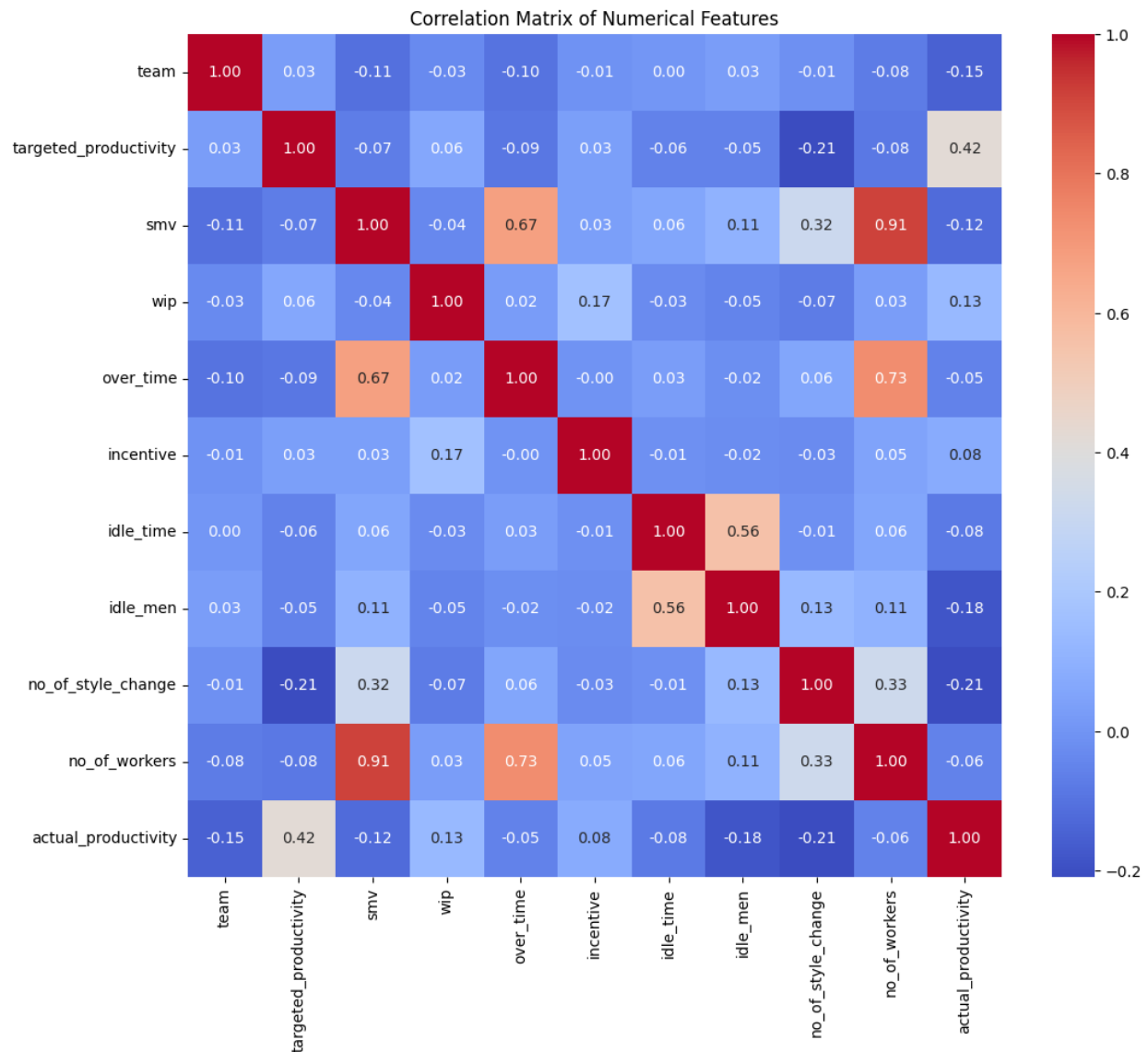
	over_time	incentive	idle_time	idle_men	\
team	-0.096737	-0.007674	0.003796	0.026974	
targeted_productivity	-0.088557	0.032768	-0.056181	-0.053818	
smv	0.674887	0.032629	0.056863	0.105901	
wip	0.022302	0.167210	-0.026299	-0.048718	
over_time	1.000000	-0.004793	0.031038	-0.017913	
incentive	-0.004793	1.000000	-0.012024	-0.021140	
idle_time	0.031038	-0.012024	1.000000	0.559146	
idle_men	-0.017913	-0.021140	0.559146	1.000000	
no_of_style_change	0.059790	-0.026607	-0.011598	0.133632	
no_of_workers	0.734164	0.049222	0.058049	0.106946	
actual_productivity	-0.054206	0.076538	-0.080851	-0.181734	

	no_of_style_change	no_of_workers	actual_productivity
team	-0.011194	-0.075113	-0.148753
targeted_productivity	-0.209294	-0.084288	0.421594
smv	0.315388	0.912176	-0.122089
wip	-0.072357	0.030383	0.131147
over_time	0.059790	0.734164	-0.054206
incentive	-0.026607	0.049222	0.076538
idle_time	-0.011598	0.058049	-0.080851
idle_men	0.133632	0.106946	-0.181734
no_of_style_change	1.000000	0.327787	-0.207366
no_of_workers	0.327787	1.000000	-0.057991
actual_productivity	-0.207366	-0.057991	1.000000

- Description: A table screenshot showing the first 5 rows of
`garments\_worker\_productivity.csv`, highlighting features like `quarter`, `smv`, and `wip`.

- Source: Generate using Pandas and Matplotlib (script provided below) or capture a screenshot
of the CSV in Excel.

2. Figure 2: Missing Values Heatmap



- Description: A heatmap visualizing missing values, emphasizing 'wip' with 42.3% missing data.

- Source: Generate using Seaborn (script provided below) or use a stock heatmap from Unsplash (search 'data heatmap').